VENKATESHWARLU SARANGI , Ph.D

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PROFESSIONAL SUMMARY

AI/ML Engineer and LLM Specialist with **10+ years of research and applied data science expertise,** holding a Ph.D. in Physics. Proficient in directing and overseeing AI projects to ensure on-time delivery, quality standards, and alignment with organizational objectives. Supervises design, development, and implementation of machine learning and deep learning models, including mining, manipulating, and statistical analysis of datasets for scalable enterprise analytics. Excels in feature selection, building and optimizing classifiers using various machine learning techniques, with a focus on best practices for model training, evaluation, and deployment emphasizing scalability, efficiency, and accuracy. Builds, mentors, and leads data science teams while collaborating with engineering, product, and business stakeholders to integrate AI solutions and translate insights into actionable strategies. Specializes in LLM fine-tuning, multimodal RAG systems, production MLOps/ LLMOps, domain-specific small-language model (SLM) optimization, Vision-Language Model (VLM) integration, and end-to-end AI architecture for fast-paced, agile environments.

TECHNICAL SKILLS

Machine Learning & Data Science: Supervised/Unsupervised Learning, Feature Engineering, Model Selection & Validation, Hyperparameter Optimization, Time Series Forecasting, Anomaly Detection, Statistical Modeling, Scikit-learn, XGBoost, LightGBM, Deep Learning Models (ResNet, DenseNet, VGG, Vision Transformers), Classifier Optimization, Linear Programming, Constraint-Based Optimization, Heuristic Methods.

RAG & Information Retrieval: Dense/Sparse Retrieval, Vector DBs (Chroma, FAISS, Pinecone, Weaviate, Milvus), Reranking & Hybrid Search, Embedding Models, Query Expansion, Document Chunking & Preprocessing, Grounding & Hallucination Detection, Evaluation Metrics.

MLOps & Infrastructure: Docker, Kubernetes (K8s), Helm; CI/CD (GitHub Actions, Jenkins, GitLab CI); Terraform; MLflow, DVC, Weights & Biases; Monitoring (Prometheus, Grafana, Evidently); Feature Stores (Feast, Tecton); A/B Testing & Experimentation; Blue-Green & Canary Deployments; Model Registry & Versioning.

Cloud Platforms: AWS (SageMaker, EC2, EKS, S3, Lambda), Azure Machine Learning, GCP Vertex AI; Serverless & Edge Deployment.

Development & Tooling: Python, PyTorch, NumPy, Pandas; FastAPI, Flask; Git, GitHub; Streamlit, Gradio (Prototyping); SQL, Spark, Airflow (Data Pipelines).

WORK EXPERIENCE

DATA SCIENTIST & ML ENGINEER

Freelancer at XRadio Ltd, HK
(Aug, 2025 – March, 2026)

Deep Learning and Multimodal Systems for ECG Signal Intelligence

- Built an explainable deep learning model for 12-lead **ECG arrhythmia classification,** addressing challenges from non-stationary signals and inter-patient variability.
- Evaluated multiple **deep learning and vision architectures** including ResNet34, ResNet50, DenseNet, and VGG16, **Vision Transformers (ViT) and Vision-Language Models (VLMs),** multimodal AI systems for healthcare prediction tasks.
- Achieved strong model performance with ResNet34, reaching **AUROC 0.98 and F1-score 0.826** across nine arrhythmia categories.

BIO-MATERIAL ENGINEER & DATA SCIENTIST

Gense Technologies Ltd, HK
(Oct, 2022 – Aug, 2025)

Biomedical Systems & Signal Intelligence

- Bio-electrodes design—Software Integration: **Architected dry electrode systems** and phantom prototypes for **EIT/ECG imaging**; **achieved 100%** improvement in contact impedance without conductive gels, directly enhancing signal fidelity and diagnostic reliability.
- Signal Processing & Enhancement: Developed and validated impedance analysis models and signal enhancement pipelines for biomedical datasets; **improved SNR and diagnostic quality** through algorithmic filtering and statistical optimization.
- Predictive Modeling: Implemented regression **(linear, polynomial, nonlinear)** and pattern extraction algorithms for predictive analytics across biomedical time-series data.
- Image Reconstruction & Validation: Built end-to-end algorithmic pipelines for **EIT image reconstruction** accuracy, applying statistical validation frameworks to quantify model performance.
- **Technologies:** Python, NumPy, Scikit-learn, Pandas; Power BI, Tableau, SQL; Impedance Analyzer.

DATA SCIENTIST AND COMPUTATIONAL ANALYST

Anvipro IT Solutions, India

(December, 2021 – June, 2022)

Materials Science & Computational Modeling

- Conducted physics-based simulation and data analysis using **COMSOL Multiphysics** to study material flexibility (>90%), stress distribution, hardness, and structural behavior.
- Performed materials **data preprocessing,** cleaning, and refinement using MATLAB and Python basic libraries to improve analysis quality and model readiness.
- Applied **supervised learning and statistical modeling** techniques to analyze material properties, identify trends, and support predictive insights.
- Improved simulation and analytical workflows through **iterative model refinement, results evaluation,** and reliability analysis for engineering and materials-focused projects.
- **Technologies:** Python, NumPy, Pandas, Scikit-learn, OriginLab, Statistical Modeling.

PH.D RESEARCH SCHOLAR

CityU HK

(August, 2015 – October 2021)

Materials Science / Computational Modeling / Data Analysis

- Designed piezoelectric electrodes, applied advanced statistical and computational modeling for material structure analysis using X-ray and neutron diffraction datasets.
- Performed **Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning** to extract structural insights from complex datasets.
- Published research in **Nature Communications, Communications Physics, Physical Review B, Scientific Reports**, and other high-impact journals (Google Scholar).
- **Technologies:** FullProf and GSAS-II for structural analysis (GUIs) and Pair Distribution Function (PDF) analysis, Python, MATLAB, OriginLab, Scikit-learn, Statistical Modeling.

JOURNAL PUBLICATIONS

- S. Venkateshwarlu et al., **Nature Communications Physics**, (2020).
- S. Venkateshwarlu et al., **Journal of Materials Research**, Cambridge University Press JMR-0830 (2019).
- S. Nayak, S. Venkateshwarlu, et al., **Journal of the American Ceramic Society**, (2019).
- Frederick, Sarangi Venkateshwarlu, et al., **Chemistry of Materials** (2021).
- A. Pramanick, S. Venkateshwarlu, et al., **Physical Review B**, (2021).
- Liang, Zhuoxin Liu, S. Venkateshwarlu, et al., **Nature**, **Light: Science & Applications**, (2021).
- G. Srinivas, S. Venkateshwarlu, et al., **Applied Physics Letters**, (2015).
- S. Venkateshwarlu, et al., **Journal of Modern Materials**, (2016).
- Pramanick, Venkateshwarlu, **Journal of the European Ceramic Society**, (2023).
- Nirmal and Sarangi Venkateshwarlu et al., **arXiv:2508.10940**, (2025).
- Kwok, W.C., Sarangi Venkateshwarlu et al., **Nature Scientific Reports** (2025).

EDUCATION | TRAINING - CERTIFICATIONS

IIT Bombay India and CityU HK,

- **Ph.D.** Materials Science and Eng (Physics), City University of Hong Kong, (Oct, 2021)
- **M.Tech.** Materials Science and Eng (Physics), IIT Bombay, India, (June, 2013)
- **Generative AI with Large Language Models (LLMs)**
- DeepLearning.AI & Amazon Web Services | Coursera | February 2026
- Skills: NLP, Transformers Architecture (FLAN-T5, BERT, GPT, DeepSeek, Qwen) · RL models · PyTorch · PEFT · Fine-tuning · LoRA · QLoRa · Knowledge Distillation · RAG · LangChain · LangGraph · Deployment · Ollama, vLLM, OpenWebUI, FastAPI, Streamlit , RunPOD, LightningAI, vLLM, HuggingFace, Optuna
- **Machine Learning Specialization** | August 2025
- **Deep Learning and NLP Specialization** | January 2026
- Stanford University & DeepLearning.AI | Coursera
- **End-to-End MLOPS Bootcamp** | Udemy | April 2026